
The Shape of a Knowledge Boundary: A Transformer Trained Only on Pre-1939 Text

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<https://github.com/Shinoaki798/1939> I used Claude Fable 5.1 to do the proposal.

Introduction

A language model’s knowledge cutoff is usually described as a wall. I suspect it is a **membrane with structure**: some post-cutoff concepts are unreachable, others are assembled from parts the model already has and are half predictable. To characterise it I will train a **decoder-only Transformer language model** from random initialisation on text ending before the Second World War — American and German press, the German in the original, because the country that started the war kept its own record — and ask three questions. **RQ1 (permeability)**: which classes of post-1939 concepts can a pre-1939 model reach? **RQ2 (perspective)**: what does reading the German press change — measured against an otherwise identical twin trained on English only? **RQ3 (foresight)**: how accurately does a model that has read everything up to June 1939 predict what happened next — the war, a third Roosevelt term, nuclear weapons — against what informed people believed at the time? Deep learning is the instrument: these quantities exist only in a trained model’s next-token distribution. Training ends **30 June 1939**; July–August 1939 is an **embargo**, excluded from training and scored evaluation.

Illustration

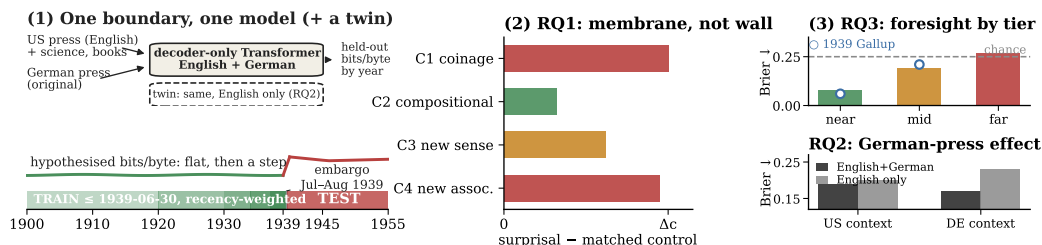


Figure 1: One boundary, one model and its English-only twin. Shapes are hypothesised, not results.

Background & Related Work

Language models degrade on text written after their training window [15]; temporal conditioning improves future-fact calibration [3]; Time Machine GPT [4] trains point-in-time models. Recent from-scratch models with 19th- and 20th-century cutoffs, including 1939 [10, 11, 20], aim to *produce* such models; this project measures the boundary. SemEval-2020 Task 1 [21], HistBERT [19] and 19th-century models [13] measure how *existing* words drift with masked encoders; RQ1 asks which *not-yet-existing* concepts a fixed earlier vocabulary can reach, which needs a causal model with a hard boundary. Gallup’s 1939 polls [7] give RQ3 a hindsight-free human reference.

Data Processing

Sources. All text is pre-cutoff, in the original language; nothing is machine-translated. *English*: American Stories [1], article-level text from Chronicling America, is the backbone and sole source of scored sets; training-only additions: Chronicling America batch OCR for the 1930s [17], the Congressional Record [8], the Federal Register [22] and case law [12] (≤10%), and public-domain books [16] (≤12%). *German*: newspapers from Europeana [6] and the Deutsches Zeitungsportal

[2], deduplicated by title and date, plus the NSDAP daily *Völkischer Beobachter* (1925, 1930) [14]; German is $\leq 25\%$ of seen tokens per period. *Science* ($\leq 10\%$, exempt from recency weighting): PNAS 1915–39 [18], the 1911 Britannica [23], the *Jahrbuch über die Fortschritte der Mathematik* [5] and Göttingen’s digitised mathematics journals [9]. **Recency weighting.** The pool centres on the 1910s but the model must live in 1939: the 1930s are seen twice; the 1920s fill the remainder ($\leq 35\%$); pre-1920 $\leq 8\%$; within a period, weight $\propto \exp(-(1939 - \text{year})/5)$. **Splits** by publication date, never shuffled: a fixed 2% of every year 1900–1955, per language, is held out and never trained on; TRAIN = the rest $\leq 1939-06-30$; VAL = a disjoint 2% $\leq$ cutoff; TEST-A/B/C = held-out 1930–38 / 1939-09–45 / 1946–55. **Cleaning**, in order: (1) language filter (stopword vote); (2) MinHash dedup of reprints (word 5-grams, Jaccard ≥ 0.8) before splitting; (3) OCR quality per language: period-lexicon hit rate ≥ 0.75 and word-like share ≥ 0.70 , ≥ 50 tokens; (4) documents containing a C1 coinage (English or German list) are dropped whole. **Format:** 48k BPE trained only on native pre-cutoff English and German; 2048-token sequences; all numbers are bits-per-byte. **RQ1 probe set:** ~ 200 post-cutoff terms in four classes (C1 coinage *radar*; C2 compositional *atomic bomb*; C3 new sense *blitz*; C4 new association *Hiroshima*), each with a frequency-matched control in ≥ 5 contexts, frozen in advance. **RQ3 propositions:** ≥ 120 binary propositions resolved after 1939-09-01, in three tiers by what was knowable in June 1939 — *near* (war in Europe this year), *mid* (a third Roosevelt term, US entry, a self-sustaining chain reaction), *far* (Hitler defeated, a fission weapon used) — each with same-topic US (English) and German (German) contexts from June and late August 1939 (embargo as conditioning only), in period language, three wordings each; continuations are English; the 1939 Gallup figure is recorded where polled.

Architecture

A **decoder-only Transformer** as in the course: token embeddings; stacked blocks of **causally masked** multi-head self-attention and position-wise feed-forward layers with residuals and layer normalisation; softmax over the vocabulary — a next-token predictor. Decoder-only because every measurement is a sequence probability of unseen text: a masked encoder’s pseudo-likelihoods give none, and an encoder–decoder has no source sequence. From scratch, no pretrained weights. **Two runs, one architecture:** the main model (English+German) and, for RQ2 only, a **twin** identical in size, tokenizer, English tokens and schedule but with the German slice removed. **Size:** 350M parameters (24 layers, $d=1024$, 16 heads, 48k vocab) on up to 10B seen tokens (≈ 29 /parameter); no text is seen more than twice (repetition raises memorisation). Compute $\approx 2.1 \times 10^{19}$ FLOPs per run: ~ 85 h on the RTX 5080 or 15 h on a rented H100. **Measurement.** RQ1: for term class c , $\Delta_c = \mathbb{E}_{t \in \mathcal{T}_c}[s(t)] - \mathbb{E}_{t \in \mathcal{C}_c}[s(t)]$, s the surprisal in fixed contexts, $\mathcal{C}_c$ matched controls, plus a C4 forced-choice test (naval station vs. attack) that doubles as a contamination check. RQ3: forced choice between two continuations, the normalised log-likelihood ratio as probability, Brier per tier, perspective and window against chance, the n -gram floor and Gallup. RQ2: the same batteries on the twin; the paired difference is the effect of the German press. All with bootstrap 95% CIs. **Success:** Δ_{C2} significantly smallest and a step at 1939 (RQ1); mid-tier Brier better than chance and not worse than Gallup (RQ3); a significant paired difference on DE-context propositions (RQ2).

Baseline Model

Following the course progression, with the same split, tokenizer and budget: a **vanilla RNN**, an **LSTM** and a **GRU** language model (2 layers, ~ 50 M parameters, next-token objective), which justify the Transformer as primary model. A **modified Kneser–Ney 5-gram** (KenLM defaults) is the non-neural floor. **Plan** (individual): corpus frozen 20 Oct; baselines 17–21 Oct; main Transformer 22 Oct–1 Nov; twin in November.

Ethical Considerations

Pre-1939 newspapers carry the racial, ethnic and gender language of their time; the model reproduces it, documented, not sanitised. German press after 1933 was state-controlled, so the “German perspective” is what Germans were permitted to read; results describe a record, not nations or culpability. In collection, OCR noise concentrates in older and German (Fraktur) text, and unreliable book dates are caught by the C1 screen. Sources are public domain or openly licensed and attributed; the model is released non-commercially, not for querying the private people the papers name.

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